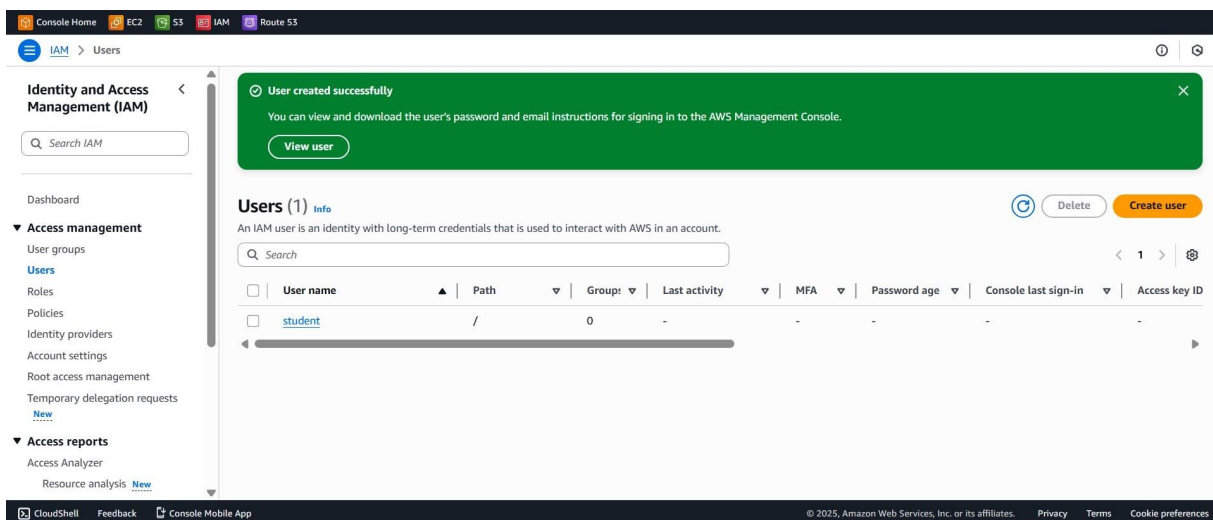


Task Description:

Create the K8s EKS, further you have to do the deployment of the Nginx application and access the application outside the cluster.

Output:

```
ubuntu@ip-172-31-44-178: ~
inflating: aws/dist/awscli/data/ac.index
inflating: aws/dist/awscli/data/metadata.json
inflating: aws/dist/awscli/data/cli.json
creating: aws/dist/awscli/customizations/sso/
creating: aws/dist/awscli/customizations/wizard/
creating: aws/dist/awscli/customizations/wizard/wizards/
creating: aws/dist/awscli/customizations/wizard/wizards/configure/
creating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/
creating: aws/dist/awscli/customizations/wizard/wizards/events/
creating: aws/dist/awscli/customizations/wizard/wizards/iam/
creating: aws/dist/awscli/customizations/wizard/wizards/lambda/
inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
inflating: aws/dist/awscli/customizations/sso/index.html
inflating: aws/dist/awscli/topics/config-vars.rst
inflating: aws/dist/awscli/topics/s3-faq.rst
inflating: aws/dist/awscli/topics/s3-config.rst
inflating: aws/dist/awscli/topics/db-expressions.rst
inflating: aws/dist/awscli/topics/return-codes.rst
inflating: aws/dist/awscli/topics/topic-tags.json
creating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
inflating: aws/dist/wheel-0.45.1.dist-info/REQUESTED
inflating: aws/dist/wheel-0.45.1.dist-info/RECORD
inflating: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt
inflating: aws/dist/wheel-0.45.1.dist-info/INSTALLER
inflating: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
inflating: aws/dist/wheel-0.45.1.dist-info/WHEEL
inflating: aws/dist/wheel-0.45.1.dist-info/direct_url.json
inflating: aws/dist/wheel-0.45.1.dist-info/METADATA
ubuntu@ip-172-31-44-178:~$ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-44-178:~$ aws --version
aws-cli/2.32.18 Python/3.13.11 Linux/6.14.0-1015-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-44-178:~$
```



Console Home EC2 S3 IAM Route 53

IAM > Users > student > Create access key

This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.

Step 1 Access key best practices & alternatives

Step 2 - optional Set description tag

Step 3 Retrieve access keys

Retrieve access keys [info](#)

Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key	Secret access key
AKIA5KQWIIWBWEXKYDFK	***** Show

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

[Download .csv file](#) [Done](#)

CloudShell Feedback Console Mobile App

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```
ubuntu@ip-172-31-44-178: ~  
creating: aws/dist/awscli/customizations/wizard/wizards/lambda/  
inflating: aws/dist/awscli/customizations/wizard/wizards/configure/ main.yml  
inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml  
inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml  
inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml  
inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml  
inflating: aws/dist/awscli/customizations/sso/index.html  
inflating: aws/dist/awscli/topics/config-vars.rst  
inflating: aws/dist/awscli/topics/s3-faq.rst  
inflating: aws/dist/awscli/topics/s3-config.rst  
inflating: aws/dist/awscli/topics/db-expressions.rst  
inflating: aws/dist/awscli/topics/return-codes.rst  
inflating: aws/dist/awscli/topics/topic-tags.json  
creating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/  
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD  
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA  
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt  
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER  
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL  
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE  
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst  
inflating: aws/dist/wheel-0.45.1.dist-info/REQUESTED  
inflating: aws/dist/wheel-0.45.1.dist-info/RECORD  
inflating: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt  
inflating: aws/dist/wheel-0.45.1.dist-info/INSTALLER  
inflating: aws/dist/wheel-0.45.1.dist-info/entry_points.txt  
inflating: aws/dist/wheel-0.45.1.dist-info/WHEEL  
inflating: aws/dist/wheel-0.45.1.dist-info/direct_url.json  
inflating: aws/dist/wheel-0.45.1.dist-info/METADATA  
ubuntu@ip-172-31-44-178:~$ sudo ./aws/install  
You can now run: /usr/local/bin/aws --version  
ubuntu@ip-172-31-44-178:~$ aws --version  
aws-cli/2.32.18 Python/3.13.11 Linux/6.14.0-1015-aws exe/x86_64.ubuntu.24  
ubuntu@ip-172-31-44-178:~$ aws configure  
AWS Access Key ID [None]: AKIA5KQWIIWBWEXKYDFK  
AWS Secret Access Key [None]: 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  
Default region name [None]: ap-south-1  
Default output format [None]: json  
AWS Access Key ID [*****YDFK]: AKIA5KQWIIWBWEXKYDFK  
AWS Secret Access Key [*****3Cse]: yes  
Default region name [ap-south-1]: yes  
Default output format [json]: yes  
ubuntu@ip-172-31-44-178:~$
```

```

ubuntu@ip-172-31-44-178: ~
inflating: aws/dist/awscli/topics/s3-config.rst
inflating: aws/dist/awscli/topics/db-expressions.rst
inflating: aws/dist/awscli/topics/return-codes.rst
inflating: aws/dist/awscli/topics/topic-tags.json
creating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
inflating: aws/dist/wheel-0.45.1.dist-info/REQUESTED
inflating: aws/dist/wheel-0.45.1.dist-info/RECORD
inflating: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt
inflating: aws/dist/wheel-0.45.1.dist-info/INSTALLER
inflating: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
inflating: aws/dist/wheel-0.45.1.dist-info/WHEEL
inflating: aws/dist/wheel-0.45.1.dist-info/direct_url.json
inflating: aws/dist/wheel-0.45.1.dist-info/METADATA
ubuntu@ip-172-31-44-178:~$ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-44-178:~$ aws --version
aws-cli/2.32.18 Python/3.13.11 Linux/6.14.0-1015-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-44-178:~$ aws configure
AWS Access Key ID [None]: AKIA5KQWIIWBWEXKYDFK
AWS Secret Access Key [None]: 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
Default region name [None]: ap-south-1
Default output format [None]: json
AWS Access Key ID [*****YDFK]: AKIA5KQWIIWBWEXKYDFK
AWS Secret Access Key [*****3Cse]: yes
Default region name [ap-south-1]: yes
Default output format [json]: yes
ubuntu@ip-172-31-44-178:~$ curl -o kubect1 https://amazon-eks.s3.us-west-2.amazonaws.com/1.29.0/2024-01-04/bin/linux/amd64/kubect1
chmod +x kubect1
sudo mv kubect1 /usr/local/bin/
kubect1 version --client
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 47.4M 100 47.4M 0 0 5457k 0 0:00:08 0:00:08 --:--:-- 7109k
Client Version: v1.29.0-eks-5e0fdde
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
ubuntu@ip-172-31-44-178:~$

```

```

ubuntu@ip-172-31-44-178: ~
creating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
inflating: aws/dist/wheel-0.45.1.dist-info/REQUESTED
inflating: aws/dist/wheel-0.45.1.dist-info/RECORD
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inflating: aws/dist/wheel-0.45.1.dist-info/INSTALLER
inflating: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
inflating: aws/dist/wheel-0.45.1.dist-info/WHEEL
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inflating: aws/dist/wheel-0.45.1.dist-info/METADATA
ubuntu@ip-172-31-44-178:~$ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-44-178:~$ aws --version
aws-cli/2.32.18 Python/3.13.11 Linux/6.14.0-1015-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-44-178:~$ aws configure
aws configure
AWS Access Key ID [None]: AKIA5KQWIIWBWEXKYDFK
AWS Secret Access Key [None]: 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
Default region name [None]: ap-south-1
Default output format [None]: json
AWS Access Key ID [*****YDFK]: AKIA5KQWIIWBWEXKYDFK
AWS Secret Access Key [*****3Cse]: yes
Default region name [ap-south-1]: yes
Default output format [json]: yes
ubuntu@ip-172-31-44-178:~$ curl -o kubect1 https://amazon-eks.s3.us-west-2.amazonaws.com/1.29.0/2024-01-04/bin/linux/amd64/kubect1
chmod +x kubect1
sudo mv kubect1 /usr/local/bin/
kubect1 version --client
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 47.4M 100 47.4M 0 0 5457k 0 0:00:08 0:00:08 --:--:-- 7109k
Client Version: v1.29.0-eks-5e0fdde
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
ubuntu@ip-172-31-44-178:~$ curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_${uname -s}_amd64.tar.gz" | tar xz
sudo mv eksctl /usr/local/bin/
eksctl version
0.220.0
ubuntu@ip-172-31-44-178:~$

```

```

ubuntu@ip-172-31-44-178:~$
2025-12-17 08:10:45 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-cluster"
2025-12-17 08:11:45 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-cluster"
2025-12-17 08:12:45 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-cluster"
2025-12-17 08:13:45 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-cluster"
2025-12-17 08:14:45 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-cluster"
2025-12-17 08:15:45 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-cluster"
2025-12-17 08:16:45 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-cluster"
2025-12-17 08:17:46 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-cluster"
2025-12-17 08:17:46 [!] recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and run 'eksctl update addon'
2025-12-17 08:17:46 [I] creating addon: vpc-cni
2025-12-17 08:17:47 [I] successfully created addon: vpc-cni
2025-12-17 08:17:47 [I] creating addon: kube-proxy
2025-12-17 08:17:47 [I] successfully created addon: kube-proxy
2025-12-17 08:17:48 [I] creating addon: coredns
2025-12-17 08:17:48 [I] successfully created addon: coredns
2025-12-17 08:17:48 [I] building managed nodegroup stack "eksctl-nginx-cluster-v2-nodegroup-linux-nodes"
2025-12-17 08:19:48 [I] deploying stack "eksctl-nginx-cluster-v2-nodegroup-linux-nodes"
2025-12-17 08:19:48 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-nodegroup-linux-nodes"
2025-12-17 08:20:18 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-nodegroup-linux-nodes"
2025-12-17 08:20:51 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-nodegroup-linux-nodes"
2025-12-17 08:22:25 [I] waiting for CloudFormation stack "eksctl-nginx-cluster-v2-nodegroup-linux-nodes"
2025-12-17 08:22:25 [I] waiting for the control plane to become ready
2025-12-17 08:22:25 [✓] saved kubeconfig as "/home/ubuntu/.kube/config"
2025-12-17 08:22:25 [I] no tasks
2025-12-17 08:22:25 [✓] all EKS cluster resources for "nginx-cluster-v2" have been created
2025-12-17 08:22:26 [I] nodegroup "linux-nodes" has 2 node(s)
2025-12-17 08:22:26 [I] node "ip-192-168-3-111.ap-south-1.compute.internal" is ready
2025-12-17 08:22:26 [I] node "ip-192-168-83-161.ap-south-1.compute.internal" is ready
2025-12-17 08:22:26 [I] waiting for at least 2 node(s) to become ready in "linux-nodes"
2025-12-17 08:22:26 [I] nodegroup "linux-nodes" has 2 node(s)
2025-12-17 08:22:26 [I] node "ip-192-168-3-111.ap-south-1.compute.internal" is ready
2025-12-17 08:22:26 [I] node "ip-192-168-83-161.ap-south-1.compute.internal" is ready
2025-12-17 08:22:26 [✓] created 1 managed nodegroup(s) in cluster "nginx-cluster-v2"
2025-12-17 08:22:26 [I] creating addon: metrics-server
2025-12-17 08:22:26 [I] successfully created addon: metrics-server
2025-12-17 08:22:27 [I] kubectrl command should work with "/home/ubuntu/.kube/config", try 'kubectrl get nodes'
2025-12-17 08:22:27 [✓] EKS cluster "nginx-cluster-v2" in "ap-south-1" region is ready
ubuntu@ip-172-31-44-178:~$ kubectrl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-3-111.ap-south-1.compute.internal Ready    <none>   108s   v1.32.9-eks-ecaa3a6
ip-192-168-83-161.ap-south-1.compute.internal Ready    <none>   107s   v1.32.9-eks-ecaa3a6
ubuntu@ip-172-31-44-178:~$

```